Business Requirements Document (BRD)

Project Name: Insurance Claims & Underwriting Risk Analytics

1. Project Background

A mid-sized insurance provider offering Health, Life, and Auto policies requires an institutional reporting solution. Historical claims and policy records were fragmented, preventing leadership from determining net underwriting risk, identifying disproportionately high-claiming customers, and assessing operational claim approval throughput across product lines.

2. Core Business Problem

- **Exposure Blindspots:** Inability to track customer claim-to-coverage exposure ratios dynamically.
- **Underwriting Friction:** Approval and rejection distributions were not tracked by demographic segments (Gender) or product lines.
- **Granular Drill-Down Limits:** Lack of a standardized enterprise date table prevented chronological and seasonal trend inspection.

3. Business Objectives & Success Criteria

Business Objective	Metric / KPI	Implementation
Top Claim Drivers	Total Claim Incurred ( )	`Total Claim Amount` measure aggregated by Customer
Operational Efficiency	Claim Approval Rate (%)	`(Approved Claims / Total Claims) * 100`
Loss & Risk Exposure	Claim-to-Coverage Ratio (%)	`DIVIDE(Total Claim Amount, Total Coverage Amount)`

Demographic Severity	Claims by Gender & Age	Dimensional slicing via `dim_customer`
Portfolio Distribution	Claims by Product Line	Multi-dimensional analysis across Health, Life, Auto

4. Solution Architecture Scope

- **Storage & Processing:** Microsoft Fabric Lakehouse (Delta Parquet tables) following Bronze -> Silver -> Gold Medallion architecture.
- **Data Modeling:** Dimensional Star Schema with 1:Many unidirectional relationships feeding a high-performance Direct Lake semantic model.
- **Visualization Layer:** Interactive Power BI executive dashboard with dynamic slicing across Year, Quarter, Gender, Claim Status, and Policy Type.